Every equation solved by `funwave_2d_sw/run_meander.py`

Nonlinear depth-averaged shallow water on a meandering channel with an erodible bed and erodible banks

`rossby_palooza/numerical/funwave_2d_sw`

2026-07-22

Abstract

This note writes down, without omission, the system that the driver `run_meander.py` sets up and that FUNWAVE-TVD integrates: the interior mass and momentum equations, every closure, how a sinuous channel is prescribed on a Cartesian grid, the boundary conditions (hydrodynamic and sedimentary), the initial condition, the bed-evolution equation, and the bank treatment. Numbers quoted are the values actually in `CONFIG`; the figures are generated from the driver itself by `make_figs.py`, so this document cannot drift from the code. How the secondary flow is parameterised is §14; the full list of approximations that bounds every result is §11.

Contents

1 Scope and sign conventions

Independent variables are the down-valley coordinate x, the cross-valley coordinate y, and time t. All fields are **depth-averaged**: there is no vertical coordinate and no vertical structure. Dependent variables:

symbol	meaning	sign
$\eta(x, y, t)$	free-surface elevation above the still-water datum	up +
$h(x, y, t)$	still-water depth	down +
$H = \eta + h$	total water depth	≥ 0
(u, v)	depth-averaged velocity	
$(P, Q) = (Hu, Hv)$	volume flux per unit width	
$\bar{c}(x, y, t)$	depth-averaged volumetric sediment concentration	
$z_b = -h$	bed elevation	up +

A trap worth stating once. FUNWAVE's internal bed-change variable `Zb` is *positive for erosion* and enters as `Depth = Depth_ini + Zb*Morph_factor`. Erosion therefore *increases* `Depth` and *decreases* z_b.

2 Interior equations

2.1 Mass

$$\partial_t \eta + \partial_x(Hu) + \partial_y(Hv) \;=\; -\,\partial_t h. \tag{1}$$

The right-hand side is the coupling to the moving bed: a rising bed displaces water. Under the morphodynamic scale separation of §10 it is dropped from the hydrodynamic step and reinstated only when the bed is updated.

2.2 Momentum

In conservative, well-balanced form — which is what FUNWAVE integrates once `DISPERSION = F` sets $\Gamma_1 = \Gamma_2 = 0$ (`src/init.F:647-652`):

$$\partial_t P + \partial_x\left[\frac{P^2}{H} + \tfrac{1}{2}g\big(\eta^2 + 2\eta h\big)\right] + \partial_y\left[\frac{PQ}{H}\right] = g\eta\,\partial_x h \;-\; C_d\,u\sqrt{u^2+v^2} \;+\; \nabla\cdot(\nu_t H\nabla u)\,, \tag{2}$$

$$\partial_t Q + \partial_x\left[\frac{PQ}{H}\right] + \partial_y\left[\frac{Q^2}{H} + \tfrac{1}{2}g\big(\eta^2 + 2\eta h\big)\right] = g\eta\,\partial_y h \;-\; C_d\,v\sqrt{u^2+v^2} \;+\; \nabla\cdot(\nu_t H\nabla v)\,. \tag{3}$$

Splitting the hydrostatic pressure as $\frac{1}{2}g(\eta^2 + 2\eta h)$ with the source $g\eta\,\nabla h$ is what makes the scheme *well-balanced*: lake-at-rest ($\eta \equiv 0$, $\boldsymbol{u} \equiv 0$) is preserved exactly over an arbitrary bed. This matters here because the bed is strongly non-flat by construction.

2.3 Closures

$$\text{bed friction (\texttt{src/sources.F:242})} \qquad \boldsymbol{\tau}_b/\rho = C_d\,\boldsymbol{u}\,|\boldsymbol{u}|, \tag{4}$$

$$\text{lateral mixing (\texttt{src/sources.F:367-385})} \qquad \nu_t = C_{\mathrm{smg}}\,\Delta x\,\Delta y\,\sqrt{u_x^2 + v_y^2 + \tfrac{1}{2}(u_y+v_x)^2} \;+\; \nu_{\mathrm{bkg}}. \tag{5}$$

with $C_{\mathrm{smg}} = 0.25$ (`src/mod_global.F:301`) and $\nu_{\mathrm{bkg}} = 0$.

C_d is not free. FUNWAVE's sediment module computes its own bed shear from a log law (§6) that is *independent* of the `Cd` in `input.txt`. Consistency requires setting

$$C_d \;=\; \frac{\kappa^2}{\left[\ln(30H/k_s) - 1\right]^2} \;=\; 0.00154 \qquad (H = 3\text{ m},\; k_s = 2.5D_{50},\; D_{50} = 0.5\text{ mm}). \tag{6}$$

Leaving the shipped example's 0.002 makes flow and transport see a bed that differs by 30%.

Relation to the linear (s, n) model. Linearising (4) about a base state $(\bar{U}, 0)$ with $H = \bar{H} + \eta'$ gives

$$\frac{C_d\,\boldsymbol{u}|\boldsymbol{u}|}{H}\bigg|_x \simeq \underbrace{\frac{C_d\bar{U}^2}{\bar{H}}}_{\text{base}} + \underbrace{\frac{2C_d\bar{U}}{\bar{H}}}_{r_s} u' - \underbrace{\frac{C_d\bar{U}^2}{\bar{H}^2}}_{r_\eta}\eta', \qquad \frac{C_d\,\boldsymbol{u}|\boldsymbol{u}|}{H}\bigg|_y \simeq \underbrace{\frac{C_d\bar{U}}{\bar{H}}}_{r_n} v'. \tag{7}$$

So the anisotropic drag $r_s = 2C_f\bar{U}/\bar{H}$, $r_n = C_f\bar{U}/\bar{H}$ and the superelevation-drag $r_\eta = C_f\bar{U}^2/\bar{H}^2$ of the (s, n) linear model are *exactly* the first-order expansion of the single term (4). The two models agree on friction to first order; FUNWAVE additionally carries all higher orders.

3 Geometry: prescribing a sinuous channel on a Cartesian grid

FUNWAVE has no curvilinear option, so the meander lives entirely in the bathymetry $h(x, y)$. Three steps.

3.1 Centreline, with a straight lead-in

$$y_c(x) \;=\; A\,T(x)\,\cos(kx), \qquad k = \frac{2\pi}{\lambda}, \tag{8}$$

where T is a raised-cosine amplitude taper that is identically zero over a straight lead-in of length ℓ_s and reaches one at the edge of the buffer ℓ_b:

$$T(x) = \tfrac{1}{2}\left[1 - \cos\!\left(\pi\,\Pi\!\left(\frac{d(x) - \ell_s}{\ell_b - \ell_s}\right)\right)\right], \qquad d(x) = \min\big(x,\; L - x\big), \tag{9}$$

Π denoting clipping to $[0,1]$. Both position and slope are continuous at the joins, and $y_c(0) = y_c(L) = 0$: **both ends of the domain sit on the valley axis, in a dead straight channel**. The reason is boundary conditions, §4: the tide sponge is 30 cells wide and the state it imposes ($u = U$, $v = 0$, η constant) is exactly correct only in a uniform straight reach.

Curvature. Because of the taper the analytic curvature of $A\cos kx$ no longer applies, so the signed curvature is taken by finite difference on the sampled curve,

$$\kappa \;=\; \frac{x'y'' - y'x''}{(x'^2 + y'^2)^{3/2}}. \tag{10}$$

In the interior $|\kappa|_{\max} \to C_0 \equiv Ak^2$; inside the sponge it is 2–8% of C_0.

3.2 Channel coordinates by nearest-point projection

For every grid point (x, y) the driver finds the nearest point on a densely sampled centreline (a KD-tree query at $\Delta s = \Delta x/4$) and returns the arc length s, the unit tangent $\hat{\boldsymbol{t}}$, the curvature κ, and the *signed* normal distance

$$n \;=\; \hat{t}_x\,(y - y_c) - \hat{t}_y\,(x - x_c), \tag{11}$$

positive to the *left* of the tangent. The first-order approximation $n \approx (y - y_c)\cos\theta$ is wrong by $O(10\%)$ at these sinuosities and is not used.

Inner versus outer bank. With (10)–(11),

$$\boxed{n\,\mathrm{sgn}(\kappa) > 0 \iff \textbf{inner bank}, \qquad n\,\mathrm{sgn}(\kappa) < 0 \iff \textbf{outer bank}.} \tag{12}$$

Using the sign of n alone would be wrong: the outer bank swaps sides every half wavelength. This is verified geometrically (centre-of-curvature distances $R \mp b$) in `tests/test_bathy.py` §7b, because getting it backwards silently inverts every conclusion about bars and scour.

3.3 Cross-section and the down-valley tilt

$$h(x, y) \;=\; \underbrace{h_{\mathrm{sec}}(n(x, y))}_{\text{shape}} \;+\; \underbrace{S\,(s(x, y) - s_0)}_{\text{tilt}}, \tag{13}$$

$$h_{\mathrm{sec}}(n) = \begin{cases} \left(w_0 + \frac{1}{2}\beta n^2\right)^{-2}, & |n| \le b, \\ \max\left(H_b - \dfrac{|n| - b}{m},\; -f\right), & |n| > b, \end{cases} \qquad w_0 = H_c^{-1/2}, \quad \beta = \frac{2\left(H_b^{-1/2} - w_0\right)}{b^2}. \tag{14}$$

Using the *arc length* s rather than x in the tilt makes the sinuosity correction $S_{\mathrm{valley}} = \sigma\, S_{\mathrm{channel}}$ automatic instead of a separate constant that can be forgotten.

Why this section shape. Under the local normal-flow balance $U = \sqrt{ghS/C_d}$ the depth-averaged relative vorticity is $\zeta = -\partial_n U$, so with $w \equiv h^{-1/2}$,

$$q \;\equiv\; \frac{\zeta}{h} \;=\; -\frac{1}{2}\sqrt{\frac{gS}{C_d}}\,\frac{\partial_n h}{h^{3/2}} \;=\; \sqrt{\frac{gS}{C_d}}\,\frac{\mathrm{d}w}{\mathrm{d}n} \quad\Longrightarrow\quad \partial_n q = \sqrt{\frac{gS}{C_d}}\,\frac{\mathrm{d}^2 w}{\mathrm{d}n^2}. \tag{15}$$

Hence w quadratic in n — which is exactly (14) — gives

$$\boxed{\partial_n q = \sqrt{gS/C_d}\;\beta \;=\; 1.105 \times 10^{-4}\ \mathrm{m^{-2}s^{-1}} \quad \text{exactly constant.}} \tag{16}$$

This is the divergent-shallow-water analogue of the constant channel-β ($\partial_n \bar{\zeta} = 2\Delta/b^2$) that the rigid-lid, streamfunction formulation imposes by prescribing a parabolic jet. Here there is no streamfunction and no prescribed jet: q is the materially conserved quantity ($f = 0$), so the constant gradient is built on q, not on ζ.

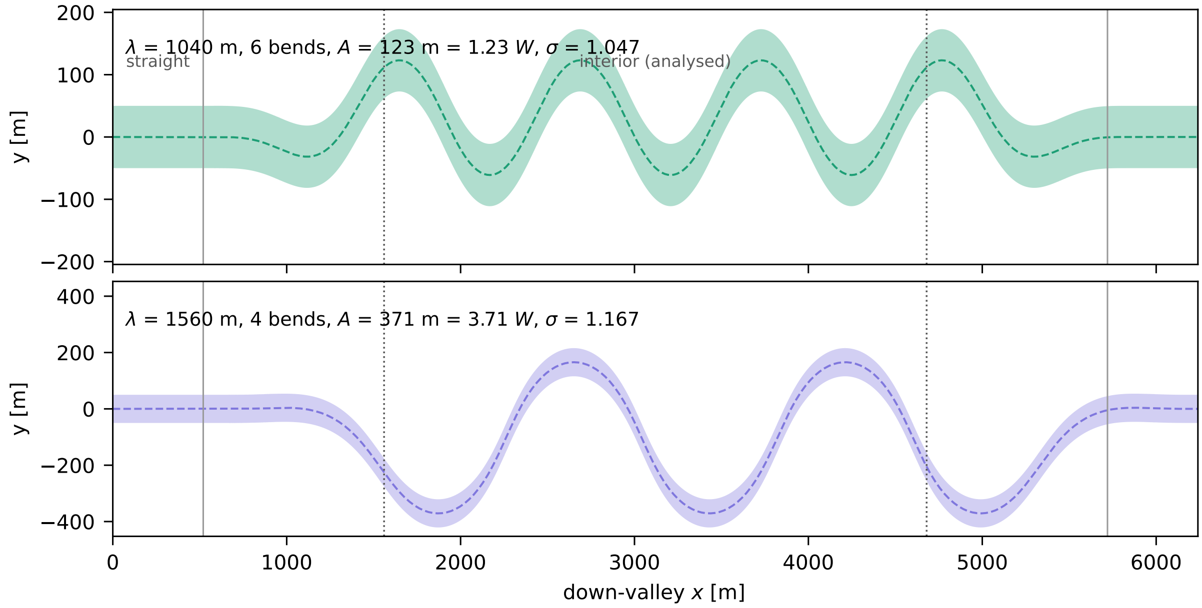

Figure 1: xOy plan view of the two reaches. Both share the down-valley length L, the grid, the buffers, and the apex curvature $C_0 = Ak^2$; only the bend *density* differs. Dotted lines: buffer edges (non-erodible); solid grey lines: end of the straight lead-in. The lateral axis is exaggerated.

The jet is an output, not a knob. Under (13) normal flow gives $U \propto \sqrt{h}$. Reproducing a prescribed excess $\Delta = 0.6\ \mathrm{m\,s^{-1}}$ over a bank speed $U_0 = 0.8$ would require $H_c/H_b = (1.4/0.8)^2 = 3.06$, i.e. a thalweg 9.2 m deep against a 3 m bank, dropping b/H from 17 to 5.4 — a different regime. A Δ-sweep at fixed geometry is therefore *not* available in this model.

4 Boundary conditions

4.1 Inflow and outflow

FUNWAVE's tide module relaxes *all three* of η, u, v toward prescribed constants inside a sponge $I_w = 30$ cells wide at each of the west and east boundaries (`src/mod_tide.F:323-330`):

$$\phi \;\leftarrow\; \phi_{\mathrm{BC}} + (\phi - \phi_{\mathrm{BC}})\,\Sigma(i), \qquad \Sigma = \left[\max\left(A_{sp}^{r}, 1\right)\right]^{-1}, \quad r = R_{sp}^{50(i-1)/(I_w-1)}, \tag{17}$$

with $R_{sp} = 0.85$, $A_{sp} = 10$. The imposed state is the uniform normal flow

$$\eta_{\mathrm{BC}} = \pm\tfrac{1}{2} S L_{\mathrm{ch}}, \qquad u_{\mathrm{BC}} = U \text{ (west)}, \qquad v_{\mathrm{BC}} = 0, \tag{18}$$

$L_{\mathrm{ch}} = \sigma L$ being the channel (not valley) length. **Every component must be given at both ends**: `TideEast_U` defaults to zero (`src/mod_tide.F:454`), which walls the outflow.

Well-posedness caveat. A subcritical open-channel reach admits *one* condition at the inlet (discharge) and *one* at the outlet (stage). Equation (17) imposes three at each. The extra conditions must therefore be made mutually consistent, which is the purpose of the calibration in §8. This is the single largest practical difficulty in driving FUNWAVE as a river: no shipped example uses steady discharge through-flow, so this operating point is untested territory for the code.

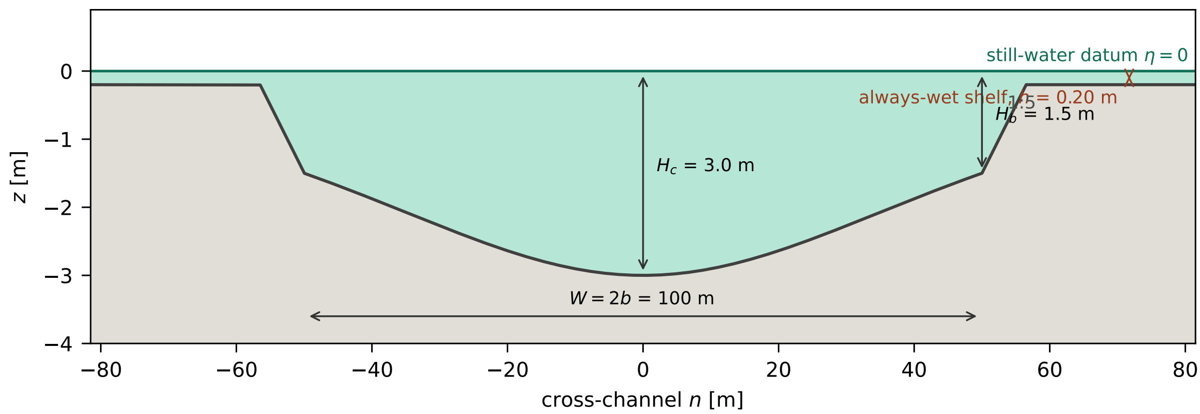

Figure 2: yOz cross-section (14), identical at every s. The constant-$\partial_n q$ bed occupies $|n| \le b$; beyond it the bank rises at 1:m to the floodplain. The gap between the still-water datum and the bank top is the freeboard analysed in §9.

4.2 Banks: a free waterline, not a wall

No boundary condition is applied at the banks. A cell is wet or dry by

$$\text{MASK} = \begin{cases} 1, & H > H_{\min} \\ 0, & H \le H_{\min} \end{cases} \qquad H_{\min} = 0.01 \text{ m}, \tag{19}$$

and fluxes through a dry face vanish. The waterline is therefore free to move up and down the bank, and **the channel width is not imposed** — in contrast with the fixed-wall condition $u_n(\pm b) = 0$ of the linear (s, n) model, which is only legitimate in the limit of vanishing bank erodibility.

The alternative that was rejected. FUNWAVE *does* support true internal vertical walls: `OBSTACLE_FILE`→`MASK_STRUC`, with `MASK_STRUC = 0` a permanent dry point ($h = -\text{LARGE}$, `src/init.F:1145-1151`). That reproduces $u_n(\pm b) = 0$ exactly and needs no elevation at all. It is not used here because `MASK_STRUC` is never updated after initialisation, so a vertical wall is necessarily a *rigid* bank — and bank erosion is the point.

4.3 Sediment

$$\text{advective scalar flux} = 0 \quad \text{on all four open boundaries} \qquad (\texttt{mod\_sediment.F:1443-1503}), \tag{20}$$

i.e. there is no equilibrium-inflow condition, and `PERIODIC` is south–north only so no streamwise-periodic reach exists. The artefact this creates is quarantined by a non-erodible buffer of *fixed physical length* ℓ_b at each end:

$$Z_s(x, y) = \begin{cases} 0, & x < \ell_b \text{ or } x > L - \ell_b \\ \infty, & \text{otherwise}, \end{cases} \qquad Z_b \le Z_s. \tag{21}$$

ℓ_b is a length and not a bend count because the artefact's scale is the sediment adaptation length $L_a = UH/(\gamma w_s) \approx 20$ m plus the flow adjustment, neither of which scales with λ.

Z_s **blocks erosion only.** It is applied as `IF(Zb>Zs) Zb=Zs` with $Z_b > 0$ meaning erosion, so deposition ($Z_b < 0$) is *not* capped and the buffer does accrete. It is a quarantine, not a frozen bed, and is excluded from analysis.

5 Initial condition

The analytic normal-flow state for the calibrated bed slope:

$$\eta_0 = -S\,(s - s_0)\,, \qquad |\boldsymbol{u}_0| = \sqrt{\frac{g\,h_{\text{sec}}\,S}{C_d}}, \qquad \boldsymbol{u}_0 = |\boldsymbol{u}_0|\,\hat{\boldsymbol{t}}, \tag{22}$$

so that $H_0 = \eta_0 + h = h_{\text{sec}}$ exactly: the initial depth is the design section everywhere, and the surface parallels the bed. Starting from rest instead would cost a full channel transit time, and because the two cases have different channel lengths that would give them *different* spin-up durations — a confound in exactly the variable under test. The run is therefore split into a rigid-bed spin-up (`Bed_Change = F`, one transit time, per-case) and a mobile-bed phase hot-started from it through `INI_UVZ` (identical duration for both cases).

6 Sediment transport

All stresses below are τ/ρ; the density is divided out of both τ_b and τ_{cr} in the source.

$$\begin{aligned}
&\text{bed shear (}\texttt{:1254}\text{)} && \tau_b = \frac{\kappa^2|\boldsymbol{u}|^2}{[\ln(30H/k_s) - 1]^2}, \qquad \kappa^2 = 0.16, \quad k_s = 2.5D_{50}, && (23)\\
&\text{critical (}\texttt{:936}\text{)} && \tau_{cr} = (s_g - 1)\,g\,D_{50}\,\theta_{cr}, && (24)\\
&\text{grain parameter (}\texttt{:935}\text{)} && D_* = D_{50}\left[\frac{(s_g - 1)g}{\nu^2}\right]^{1/3}, && (25)\\
&\text{settling (Soulsby 1997)} && w_s = \frac{\nu}{D_{50}}\left[\sqrt{10.36^2 + 1.049D_*^3} - 10.36\right] = 0.0745\ \text{m}\,\text{s}^{-1}. && (26)
\end{aligned}$$

6.1 Erosion (pickup) — van Rijn (1984), `:1287--1297`

For $\tau_b > \tau_{cr}$ and $H > H_{\text{pick}}$:

$$c_b = 0.015\left[\frac{\tau_b - \tau_{cr}}{\tau_{cr}}\right]^{3/2} D_*^{-0.3}, \qquad c_a = \min\left(1,\ \frac{0.65}{c_b}\right) c_b \frac{D_{50}}{0.01H}, \qquad \boxed{P = \max\left(0,\ c_a w_s\right).} \tag{27}$$

H_{pick} is load-bearing. The shipped example uses $H_{\text{pick}} = 0.1$ m, which switches pickup off in shallow water — that is, *at the bank toe*, which is the entire bank-retreat mechanism. It cannot be zero either: the log law (23) is singular at $H = e\,k_s/30 = 1.13 \times 10^{-4}$ m. We use $H_{\text{pick}} = 0.01$ m, 88× above the singularity and 10× below the example.

6.2 Deposition — Cao, `:1369--1372`

$$\gamma = \min\left[2,\ \frac{1 - n_p}{\bar{c}}\right], \qquad \boxed{D = \gamma\,\bar{c}\,w_s\,(1 - \gamma\bar{c})^2\,.} \tag{28}$$

Erosion and deposition are both always on. P depends on τ_b and *not* on $\bar{c}$ (a pure source); D is linear in $\bar{c}$ (a sink). They act simultaneously at every wet point. The bed responds to the *net* $D - P - \nabla\cdot\boldsymbol{q}_b$; under capacity transport $D = P$ and the bed does not move. One cannot have erosion without deposition: (29) would then have no sink and $\bar{c}$ would grow without bound. Deposition is also what builds bars, so removing it would remove the phenomenon of interest.

6.3 Suspended load

$$\partial_t(\bar{c}H) + \nabla\cdot(\bar{c}H\boldsymbol{u}) = \nabla\cdot(kH\nabla\bar{c}) + P - D. \tag{29}$$

Because P is $\bar{c}$-independent and D is linear in $\bar{c}$, (29) is a relaxation equation with time and length scales

$$T_c = \frac{H}{\gamma w_s} = 20.1\ \text{s}, \qquad L_a = U\,T_c = 20\ \text{m} = 0.20\,W. \tag{30}$$

6.4 Bedload — Meyer-Peter & Müller (1948), `:1319--1323`

For $\tau_b > \tau^b_{cr}$:

$$|\boldsymbol{q}_b| = \frac{8\left(\tau_b - \tau^b_{cr}\right)^{3/2}}{g\,(s_g - 1)}, \qquad \boldsymbol{q}_b = |\boldsymbol{q}_b|\,\big(\cos\alpha, \sin\alpha\big), \qquad \alpha = \operatorname{atan2}(v, u). \tag{31}$$

7 Bed and bank evolution

7.1 Exner

$$\boxed{(1 - n_p)\,\partial_t z_b \;=\; D - P - \nabla\cdot\boldsymbol{q}_b} \tag{32}$$

implemented in cumulative form with $Z_b > 0$ meaning erosion (`:1531--1546`):

$$Z_b = -\frac{1}{1 - n_p}\left[\int \left(\bar{D} - \bar{P}\right)\mathrm{d}t \;-\; \int \nabla\cdot\boldsymbol{q}_b\,\mathrm{d}t\right], \qquad Z_b \le Z_s, \tag{33}$$

where $\bar{P}, \bar{D}$ are means over the window `Morph_interval`. The bed seen by the hydrodynamics is then

$$h = h_{\text{ini}} + Z_b \cdot \text{MF}, \qquad \text{MF} \in \mathbb{Z}\ (\texttt{Morph\_factor}). \tag{34}$$

7.2 Banks

There is **no separate bank law**. A bank is an ordinary bed cell at the channel margin and erodes through (27)–(32) like any other. Bank *retreat* therefore has two ingredients:

1. **toe erosion** — (32) steepens the bank face;
2. **geotechnical failure** — wherever the local slope exceeds the angle of repose, material is moved down-slope (`:1554--1637`), every `Aval_interval`:

$$\text{if} \quad \max_{j\in\mathcal{N}(i)} \frac{z_{b,i} - z_{b,j}}{\Delta x} > \tan\phi \quad \text{and} \quad Z_{b,i} < Z_{s,i}: \quad \delta = \tfrac{1}{2}\,(z_{b,i} - z_{b,j}) - \tfrac{1}{2}\tan\phi\,\Delta x, \tag{35}$$

   with $z_{b,i} \mathrel{-}= \delta$, $z_{b,j} \mathrel{+}= \delta$.

This *derives* bank retreat rather than prescribing it, but its rate is set by $D_{50}, \theta_{cr}, \tan\phi$ — **not** by an erodibility coefficient E. It is therefore not comparable, without recalibration, with a kinematic law $\partial_t\zeta_c = E \cdot \frac{1}{2}[u'_s(+b) - u'_s(-b)]$.

8 Head budget and its calibration

The straight-channel normal slope $S_0 = C_d U^2/(gH_c)$ omits bend losses. Imposing it together with the inlet velocity over-specifies the problem: the reach cannot pass the design discharge on that head, draws down, loses conveyance, and diverges. The bed slope and the boundary head are therefore scaled together by a calibrated factor,

$$S = \mathcal{H}\,S_0, \qquad \mathcal{H}_{\text{new}} = \mathcal{H}\left(\frac{U}{U_{\text{meas}}}\right)^2, \tag{36}$$

the update following from $U \propto \sqrt{S}$. Scaling *both* keeps the state self-consistent ($H = h_{\text{sec}}$ at uniform flow). Measured: $U \propto \sqrt{S}$ holds to 0.8%, and the reach delivers $\approx 82\%$ of the straight-channel target, i.e. $\mathcal{H} \approx 1.5$ at $R/W = 2$.

9 Will the water leave the channel?

No. The free surface is bounded by the imposed head drop plus the bend superelevation:

$$\boxed{\eta_{\max} \simeq \tfrac{1}{2} S\, L_{\text{ch}} + \frac{U^2 W}{2gR} < f \quad \text{(bank top)}.} \tag{37}$$

For the present configuration:

case	$\frac{1}{2}SL_{\text{ch}}$	superelevation /2	$\eta_{\max}$	margin to $f = 1.5$ m
$\lambda = 1040$ m	0.154 m	0.018 m	0.172 m	1.328 m (8.7×)
$\lambda = 520$ m	0.129 m	0.018 m	0.147 m	1.353 m (10.2×)

The flow is confined to the channel and erosion is restricted to the bed and the banks, as intended. Note that the observed failure mode in the unstable configurations is the *opposite* of overflow: the reach *drains* ($\eta \to -73$ m in one case), which FUNWAVE does not flag because its blow-up threshold is 100× the maximum depth. Run health must therefore be judged on the fields — fraction of the channel still wet, and $|\eta|$ inside the channel — never on the string `Normal Termination`.

10 Timescales

$$T_{\text{flow}} = \frac{\text{CFL}\,\Delta x}{U + \sqrt{gH}}, \qquad T_c = \frac{H}{\gamma w_s}, \qquad T_{\text{bed}} = \frac{(1-n_p)H\,\mathcal{L}}{q_b}, \qquad T_{\text{bank}} = \frac{b}{\varepsilon U}. \tag{38}$$

At $U = 0.85\ \text{m}\,\text{s}^{-1}$, $D_{50} = 0.5$ mm:

	T_{flow}	T_c	T_{bed} (one bar)	T_{bank}
value	0.20 s	20.1 s	1.68×10^7 s = 194 d	$\sim 5 \times 10^9$ s

Two consequences. (i) `Morph_interval` must exceed T_c by a good margin or the Exner forcing is an unequilibrated suspension transient; we use 200 s $\approx 10\,T_c$. (ii) $T_{\text{bed}}/T_{\text{flow}} \sim 10^8$, bridged by MF = 1000 in (34); T_{bank} is a further factor beyond and is *not* reachable, so this model gives bar and bend morphodynamics, not century-scale planform migration.

11 Complete list of approximations

This model is **not** a first-principles simulation. Every approximation it makes is listed here; results can only be read through this list.

11.1 Physical

1. **Hydrostatic pressure**, $p = \rho g(\eta - z)$: the long-wave limit $L \gg H$. Vertical accelerations are neglected.
2. **Depth-averaged**: no vertical coordinate, so secondary (helical) circulation cannot be *resolved*. It is instead *parameterised* through the two Ikeda closures of §14 — drag modulation (40) and bedload deflection (41) — rather than being absent ($A = 2.89$, not $A = 0$).

3. **Non-dispersive**: `DISPERSION = F` drops all Boussinesq terms ($\Gamma_1 = \Gamma_2 = 0$). Valid because $kH \ll 1$ for a river.
4. **Non-rotating**, $f = 0$. Any "Rossby wave" here is a shear-β vortical wave, never a planetary one.
5. **Uniform, non-cohesive grain size** D_{50}: no grading, no armouring, no cohesion, no bedforms (ripples/dunes are inside $k_s = 2.5D_{50}$, not resolved).
6. **Empirical transport closures**: log-law drag (23), van Rijn pickup (27), Cao deposition (28), Meyer-Peter–Müller bedload (31), Soulsby settling. Each carries its own calibration range; none is derived.
7. **Bedload direction** = depth-averaged flow direction, *plus* the secondary-flow deflection (41) toward the inner bank and the gravitational return down the transverse slope (42); their balance is Ikeda's equilibrium tilt (§14).
8. **Smagorinsky eddy viscosity** (5) as the sole lateral closure.
9. **Avalanching is instantaneous** relaxation to $\tan\phi$ once per `Aval_interval`; no soil mechanics, no pore pressure, no cohesion.
10. **Rigid, impermeable floodplain**: no infiltration, no overbank sediment.

11.2 Timescale separation — the model is *not* fully coupled

This is the largest approximation and it is deliberate.

11. **Morphological acceleration**, MF = 5 in (34). The bed is advanced five times faster than the flow that drives it. Formally this assumes the flow is slaved to the bed — valid when the bed moves negligibly over a hydrodynamic adjustment time, which §10 supports ($T_{\text{bed}}/T_{\text{flow}} \sim 10^8$) but which **must be verified by an MF-convergence sweep**, not assumed. (MF = 50 blew up on the flow-driven inner-scour hole before the closures of §14 redirected it.)
12. **Time-averaged Exner forcing**: P and D are averaged over `Morph_interval` = 200 s $\approx 10\,T_c$ before they move the bed (32). This is exactly the "average over the fast timescale, then erode" step; it filters suspension transients out of the bed forcing.
13. **$\partial_t h$ dropped from the hydrodynamic mass equation** (1): the flow does not see the bed moving *within* a hydrodynamic step, only between bed updates.
14. **Rigid-bed spin-up**: phase 1 runs with `Bed_Change = F`, i.e. the bed and banks are literally rigid on the fast timescale, and the mobile-bed phase is hot-started from the converged flow.

Items 11–14 together *are* a fast/slow decomposition. A stronger, fully decoupled alternative is discussed in §12.

11.3 Numerical

15. **Finite resolution** $\Delta x = \Delta y = 2.5$ m: 40 cells across the channel, 6 across the bank face. The bank is a staircase at this resolution.
16. **MUSCL–TVD reconstruction with an HLL Riemann solver**, third order in space; the limiter is diffusive at extrema.
17. **Wetting–drying threshold** $H_{\min} = 0.01$ m and the friction floor H_{frc}; a cell below threshold is removed from the solution and its water is booked into a mass-adjustment term.
18. **Pickup floor** $H_{\text{pick}} = 0.01$ m: no entrainment in water shallower than this, a numerical necessity because (23) is singular at $H = e\,k_s/30$.

19. **Sediment is closed at the open boundaries**, worked around by the non-erodible buffer (21); the buffer is excluded from analysis.
20. **The boundary condition is over-specified** (§4) and made consistent only by the head calibration (36), which is itself an iterative, empirical procedure.
21. **Geometry is modified inside the buffer** by the amplitude taper (9): the first and last ℓ_b of the reach are not the meander being studied.

12 A fully decoupled alternative

Items 11–14 accelerate the bed while leaving the two systems dynamically coupled at every step. In the strongly separated regime of §10 a cleaner scheme is available, in which the fast problem is solved to *steady state* before the slow one is touched:

(i) freeze the bed; integrate (1)–(3) to a steady state $(\boldsymbol{u}^*, \eta^*)$;
(ii) time-average over the last several $T_c \Rightarrow \langle \boldsymbol{u} \rangle, \langle \eta \rangle$;
(iii) evaluate $P, D, \boldsymbol{q}_b$ from $\langle \boldsymbol{u} \rangle$ only;
(iv) advance (32) by one explicit morphological step Δt_m;
(v) rebuild h and return to (i).

This is more rigorous than (34) in three ways: the sediment is computed from a genuinely steady flow rather than from a partially-adjusted one; Δt_m is explicit and its convergence directly testable; and no feedback instability can arise from the bed outrunning the flow. Its cost is one flow re-convergence per morphological step, which from a hot start is short.

It also unlocks a diagnostic that the accelerated scheme cannot provide — see §13.

13 Can a dispersion relation be extracted?

Not from a nonlinear initial-value run. Eigenmodes, $\sigma(k)$ and $c(k)$ are properties of a *linear operator*. Equations (1)–(3) are integrated nonlinearly, superposition does not hold, and a single run therefore contains no dispersion relation. The linear (s, n) package obtains one because it linearises about a frozen base state; that route is closed here by construction.

The hydrodynamic question is already settled, and negatively. Whether the meander is a gravity wave or a vortical (shear-β) wave was tested in the linear package and the vortical claim was *retracted*: every growing run had $T_{\text{shear}} \leq 0$, i.e. the growing disturbance *loses* energy to the mean flow. Moreover the Froude test is analytically null — writing $P \equiv g\eta$, the momentum equations contain no F and F survives only as F^2 multiplying the continuity time derivative, so F-insensitivity confirms the rigid-lid reduction rather than discriminating wave families. A nonlinear solver cannot improve on either result.

What *can* be measured, and is worth measuring. The productive question is not gravity-versus-Rossby but *free morphodynamic (alternate-bar) mode versus forced boundary mode*, and for that a dispersion relation can be built numerically from the decoupled scheme of §12 by finite-differencing the nonlinear operator:

1. start from a straight channel at equilibrium;
2. perturb the bed by a small sinusoid, $z_b \to z_b + \epsilon \sin(kx)\,\Phi(n)$, with $\epsilon \ll H$;
3. run step (i)–(iii) of §12 to get $\nabla\cdot\boldsymbol{q}_b$ from the re-converged steady flow;

4. project the response onto mode k and form

$$\sigma(k) \;=\; -\frac{1}{(1-n_p)\,\epsilon}\,\langle \nabla\cdot \boldsymbol{q}_b - D + P,\ \sin(kx)\Phi(n)\rangle\,, \tag{39}$$

   the growth rate of the morphodynamic mode; the quadrature phase gives the migration speed $c(k)$;

5. repeat over k and over ϵ to confirm linearity.

This is a genuine, numerically-obtained dispersion relation for the *bed*, which is the quantity that bar theory predicts and that selects λ. It requires the closures of §14: with $A = 0$ (stock FUNWAVE) and no slope deflection there is no free bar mode to find, so (42)–(41) are a prerequisite — now implemented — not an optional refinement.

14 The secondary-flow closures (gap 1)

A depth-averaged model has no vertical coordinate, so the helical secondary flow of a bend cannot arise on its own. Its two morphological effects are therefore *parameterised* following Ikeda, Parker & Sawai (1981), and both are now active (stock FUNWAVE is the $A = 0$ "incised" limit; the alluvial value is $A = 2.89$, Suga 1963). Together they are what turns the free-vortex *inner*-bank scour of a bare depth-averaged model into the outer-bank scour and inner point bar of a real alluvial river.

Momentum half (Ikeda eq. 3b), `run_meander.py`, `FRICTION_MATRIX`. The tangential-momentum balance $U\partial_s u' = -g\partial_s \xi' - (C_f U^2/H)(2u'/U - \xi'/H + \eta'/H)$ carries the secondary flow only through the last friction term, closed by the equilibrium bed tilt $\eta'/H = -A\,\mathcal{C}\,n$. FUNWAVE already solves the other two terms (its $-C_d u|u|$ *is* $2u'/U$; its $-g\,\partial_s\eta$ *is* $-\xi'/H$), so only η'/H is added — as a spatial modulation of the drag,

$$C_d^{\mathrm{eff}}(x,y) = C_d\big(1 + A\,\kappa\,n\big), \tag{40}$$

in the code's (κ, n) sign convention (outer bank $\kappa n < 0$, verified against the centre-of-curvature distance in `tests/test_bathy.py` 7b). There $C_d^{\mathrm{eff}} < C_d$, the flow accelerates outward, and the free-vortex inner-fast tendency is opposed. This term is *weak* at $R/W \sim 3$: the friction-adjustment/wavelength factor $k_f/\sqrt{k_f^2 + k_\lambda^2} \approx 0.17$ (with $k_f = 2C_f/H$, $k_\lambda = 2\pi/\lambda$) caps the redistribution, so the steady outer/inner apex speed ratio moves only $0.63 \to 0.66$ — it reduces but does not reverse the free vortex. The bar therefore needs the bedload half.

Bedload half (Ikeda eq. 6) — the bar-building term, `mod_sediment.F`. The point bar is a *bedload* feature. The secondary flow rotates the near-bed velocity, hence the bedload vector, toward the inner bank by

$$\delta = \frac{A\,\kappa\,H}{f(\theta)}, \qquad f(\theta) = A_{\mathrm{bl}}\sqrt{\theta}, \quad \theta = \frac{\tau_b}{(s-1)gD_{50}}, \tag{41}$$

added to the flow direction, with $\kappa(x,y)$ read from a curvature field. Balanced against the transverse bed-slope (Talmon 1995) return flux

$$q_{b,n} = |\boldsymbol{q}_b|\Big[\sin\delta - \tfrac{1}{f(\theta)}\partial_n z_b\Big], \tag{42}$$

the transverse balance $q_{b,n} = 0$ gives $\partial_n z_b = -A\,\kappa\,H$ — Ikeda's equilibrium tilt (deep OUTER, shallow INNER), *independent of* $f(\theta)$, which cancels. Because it is a forcing (not diffusion), the deflection builds the bar from a flat bed, where the Talmon term alone would only diffuse toward flat; the Talmon term is retained as the down-slope stabiliser that arrests the scour hole.

`postprocessing/01_validate.py` reports the measured inner/outer bed-change split, and `04_xsection.py` shows the transverse profile evolving. The prediction these closures make is outer scour *with* an inner point bar; the stock ($A = 0$) model scours the inner bank instead.

Still absent: any resolved vertical structure; rotation ($f = 0$, so "Rossby wave" here is a shear-β vortical wave, never a planetary one); along-channel depth variation in the initial bed (any $h(x)$ must *emerge* from (32)); and width variation as an independent parameter.